MILITARY SPECIFICATION

CLOCK, ELECTRIC, INTERNALLY-ILLUMINATED, WALL-MOUNTED

1. SCOPE

1.1 This specification covers a synchronous, motor-driven electric wall clock featuring controllable, internal red illumination. The clock is intended primarily for shipboard use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

J-C-90 - Cable, Power, Electrical (Flexible Cord and Flexible Wire).
L-A-499 - Allyl Plastic Sheets, Rods, and Tubes.
L-M-525 - Molding Plastic, Polystrene, Modified.

MILITARY

MIL-T-27 - Transformers and Inductors, (Audio, Power, and Pulse).
MIL-C-3767 - Connectors, Electrical (Power, Bladed Type) General Specification for.
MIL-Q-9858 - Quality Control Systems Requirements.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

BUREAU OF SHIPS

9000-S6204-73687 - Dial Markings for IC Telegraph and Indicating Systems.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bids shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Threads Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

UNDERWRITERS' LABORATORIES, INCORPORATED.
Subject No. 863 - Standard for Time Indicating and Recording Appliances

(Copies of Underwriters' Laboratories, Incorporated, standards may be obtained from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, New York; 207 East Ohio Street, Chicago 11, Illinois; or 1655 Scott Lane, Santa Clara, California.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Preproduction model.- Electric clocks furnished under this specification shall be a product that has been tested and successfully passed the preproduction inspection specified herein. Preproduction models shall be equivalent in all respects to the production units which the contractor proposes to furnish under this specification. Prior to submission of any production units for quality conformance inspection, the contractor shall furnish a specified quantity (see 6.1) of preproduction models for examination and test to determine conformance with this specification. If the first submission of the preproduction models fails to meet this specification, the deficiencies shall be corrected and the models resubmitted for approval.

3.2 General requirements.- The clock shall be driven by a self-starting synchronous motor. The clock dial shall be 12 inches in diameter, and shall indicate a total of 12 hours in one revolution of the hour hand. The dial markings, hour, minute, and sweep second hands shall be internally red-illuminated using electro-luminescent lighting (see 3.4.6). Illumination shall be controllable through a dimming rheostat and control knob. The clock shall be designed for three-point wall mounting. The clock shall be capable

Source: https://assist.dla.mil -- Downloaded: 202610-19T16:27Z
Check the source to verify that this is the current version before use.

of synchronous operation under the service conditions specified in 3.5.

3.2.1 Fire and casualty hazards.-

3.2.1.1 The supplier shall submit to the procuring activity verification that the clock conforms to the Standard for Time-Indicating and Recording Appliance, UL 863 of the Underwriters' Laboratories, Inc. The label or listing of the Underwriters' Laboratories, Inc., will be accepted as conforming to this requirement.

3.2.1.2 In lieu of the label or listing, the supplier may submit independent proof including actual test results, satisfactory to the procuring activity that his product conforms to the published standard including methods of tests of the Underwriters' Laboratories, Incorporated.

3.2.1.3 Compliance with the above preliminary requirements regarding fire and casualty hazards does not absolve the supplier from complete compliance with the other requirements of this specification.

3.3 Material.- All metals used shall be entirely suitable for the purpose. They shall be of suitable corrosion-resisting materials or other material treated in a manner to render them adequately resistant to corrosion. Contact of dissimilar metals which may cause electrolytic action shall be avoided. Protective coatings that will crack, chip, or scale under normal service or under the test conditions of this specification shall be avoided. Aluminum alloy parts shall be anodized.

3.4 Construction.-

3.4.1 General construction.- The clocks shall be dusttight and so constructed that no parts are likely to work loose in service. The strains, impacts, shocks, vibrations, and such other conditions incident to shipping, storage, installation, and service use shall not affect their serviceability or accuracy. The construction shall be such as to withstand the service conditions specified in 3.5.

3.4.2 All screws, nuts and threaded parts shall conform to American National Standards in accordance with Handbook H28. All screws or nuts shall be adequately secured with lockwashers, elastic stop nuts, or similar devices to prevent loosening when subjected to vibrations.

3.4.3 Case.- The clock case shall be made of spun drawn, formed, or cast aluminum of a commercial grade normally employed for such purposes having a minimum thickness of 1/32 inch or molded of type III plastic in accordance with L-M-526. Threaded metal inserts shall be integrally molded in the plastic case, as required, to receive all threaded fasteners. The metal case shall be black anodized. Molded plastic cases shall be supplied in standard black mold finish. The case shall be 13 inches plus or minus 1/2 inch in diameter, and shall have an inside depth of not less than 2 inches nor more than 4 inches. Suitable openings in the side of the case shall be provided for the manual setting control, the dimming rheostat control, and the power cord. The setting control shall be located at the numeral "7" position; the dimming rheostat control shall be located at the numeral "5" position, and the power cord shall be located at either the "28" or "32" minute position.

3.4.4 Case back.- The case back shall be made of natural anodized aluminum having a minimum thickness of 1/16 inch. It shall be secured to the metal case by means of machine screws or threaded into metal inserts molded as an integral part of the plastic case. The back shall contain three mounting brackets located at numeral "10", "2", and "6" positions. Each bracket shall contain a 1/4 inch diameter hole (see 3.5.1). The brackets shall be black anodized. A warning label, decal, or metal plate shall be secured to the rear of the case back and shall contain the following:

"CAUTION HIGH VOLTAGE"
"Pull plug from electrical outlet before removing back of clock. If clock arrives with crystal broken, DO NOT connect to power source. If crystal breaks in service, pull plug before handling clock."

3.4.5 Dial cover.- The dial cover shall be made of convex glass, suitable for the application, or of plastic conforming to L-A-499. The dial cover shall be clear and free from blemishes or imperfections which would interfere with the readability of the dial. It shall be securely fastened to the case, yet shall be capable of replacement in case of damage. The dial cover shall not become damaged when subjected to the service conditions specified in 3.5.

3.4.6 Illumination.-

3.4.6.1 The clock shall be internally red illuminated suitable for minimum detrimental effect on dark adaptation by electro-luminescent lamps. The dial, hour, minute and sweep second hands, shall each be an electro-luminescent lamp, all energized from the same power source. Lamps shall be so designed that initial red illumination shall be not less than 7.0 footlamberts when the dimming rheostat control is set at the maximum brightness position. After two years of continuous operation at rated voltage (600 volts), the brightness value shall not be less than 0.5 footlamberts. Variation in brightness within each lamp or among all lamps shall not exceed a ratio of 3:1 at any brightness level selected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:27Z
Check the source to verify that this is the current version before use.

Lamps shall be sealed with a moisture-resistant coating, and shall be insulated to avoid electric shock hazard. Protective devices such as fuses and current limiting resistors shall be provided as required. Slip rings shall be employed for transmitting electrical energy to the hour, minute, and sweep second hands. Slip rings shall operate satisfactorily in accordance with this specification, shall make firm, low-resistance non-radio interference contact, and shall retain their setting in any position of the equipment and under any service condition (see 3.5). The contacts shall be noncorrosive and shall be readily accessible for inspection, cleaning, adjustment or replacement.

3.4.6.2 Dial.- The dial shall be 12 inches ± 1/4 inch in diameter. It shall be flat and shall have a conventional 12-hour clock presentation with an open minute track. The minute track shall be not less than 11 inches in diameter, graduated every minute with heavier markings at each 5-minute position. Each minute position shall be least 3/32 inch in width by 5/16 inch in length, and each 5-minute position shall be at least 3/16 inch in width by 1/2 inch in length. Numerals 1 through 12 shall be at least 1-1/4 inches in height and shall be located inside the minute track adjacent to each 5-minute position. The height to width ratio of numerals and letters shall be in accordance with Sheet 7 of Drawing 9000-S6504-73687.

The following information shall be located below the center of the dial and above the numeral "6":

Manufacturer's name
U. S. Navy
(1/8 inch letter height)

The numerals, minute markings, and nomenclature information shall be painted with red pigmented fluorescent paint to provide red illumination when the electroluminescent lamp is energized. The dial surface, except as otherwise specified herein, shall be painted a dull, nonreflecting black.

3.4.6.3 Hands.- The hour, minute, and sweep second hands shall be of conventional size and shape. The top surface of all hands shall be painted with red pigmented fluorescent paint to provide red illumination when the electroluminescent lamps are energized. The minute and sweep second hands shall be of sufficient length to coincide with the outside diameter of the minute track.

3.4.7 Mechanical components.-

3.4.7.1 Movement.- The movement shall be first grade both as to materials and workmanship. Gears shall be of clock brass or of other material of equal or superior wearing quality suitable for the purpose intended. Fiber gears will not be acceptable. All gears, pivots, and pinions shall be cleancut, smoothly finished, and free from burrs. Special provision shall be made for the self-lubrication of the more rapidly moving parts to insure long life and freedom from frequent servicing. The lubricant shall be of such quality as to insure synchronous operation at all temperatures from 40 degrees Fahrenheit (°F). to 149° F. All components shall operate without objectionable noise or hum.

3.4.7.2 Setting device.- The device for manual setting the minute hand shall project from the case at the numeral "7" position. It shall be of the pull-down and reset type.

3.4.8 Electrical components.-

3.4.8.1 All electrical components shall be so designed and constructed as to comply with Underwriters Laboratories, Inc., publication UL No. 863 for Time Indicating and Recording Appliances.

3.4.8.2 Motor.- The motor shall be of the synchronous self-starting type for use on 115 volts, 60 cycles, single-phase alternating current (A.C.). The motor shall be fully capable of instantaneous starting of the clock and driving the movement and hands while maintaining synchronous speed when connected to a 60-cycle, 108-volt source of power (see 4.5.1). When connected to 115 volts, 60 cycles, the power required to operate the motor, movement, and hands shall not exceed 5 watts.

3.4.8.3 Cable and plug.- Six feet of three wire flexible cable, conforming to type V or VI, size 18 of J-C-90 and a plug connector type UP 121M of MIL-C-3767 shall be furnished with each clock. The cable shall be supported inside of the clock case in such a manner that no mechanical stress can be transmitted to the electrical connection. The cable shall be protected against abrasion or wear at point of entry into the case by means of a rubber grommet.

3.4.8.4 Transformer.- The transformer used to step-up the voltage for energizing the electroluminescent lamps shall be in accordance with MIL-T-27. The transformer shall have a minimum life expectancy of 10,000 hours and a secondary voltage not to exceed 600 volts.

3.4.8.5 Dimming rheostat control.- A dimming rheostat control shall be provided within the case and operable from without by means of a knob located on the side of the case at the numeral "5" position. Means shall be provided positively to secure the rheostat to the case to prevent rotation and damage to the electrical connections. The rheostat shall be connected to the primary side of the step-up transformer. The rheostat shall provide uniform dimming of all lamps from maximum brightness down to 0.05 foot-lamberts before the control reaches the "off" position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:27Z
Check the source to verify that this is the current version before use.

3.5 Service conditions.- The materials and construction shall be such that each clock will operate under the following service conditions and shall withstand the tests of 4.5:

Vibration
Temperature

3.5.1 Each clock shall be provided with a suitable shock and vibration mounting attached to the case brackets. Construction shall be either individual shock mountings attached to the holes in the case brackets, or as a separate one-piece mounting with provision in either case of mounting to a vertical surface with 1/4 inch diameter machine screws.

3.6 Instruction sheet.- An instruction sheet shall be furnished with each clock. It shall include a brief general description of the assembled clock power requirements, safety features, operating controls, mounting information, and a statement regarding care to be exercised in handling the clock particularly in respect to presence of high voltage. In addition, a schematic wiring diagram shall be furnished.

3.7 Workmanship.- All parts shall be manufactured and finished in a thoroughly workmanlike manner and in accordance with the best commercial practice for the manufacture of this type of equipment. All parts shall fit and operate in a first class workmanlike manner.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality control system.- The contractor shall provide and maintain a quality control system acceptable to the Government for supplies covered by this specification. The system of quality control shall be in accordance with MIL-Q-9858.

4.3 Preproduction inspection.- The preproduction model (see 3.1) shall be examined and tested to determine compliance with all requirements of this specification. All examinations and tests specified in 4.5 and 4.6 shall be conducted. Approval of the preproduction model by the procuring activity concerned shall not relieve the contractor of his obligation to supply equipment conforming to this specification. Any changes or deviations shall be subject to the approval in writing of the contracting officer.

4.4 Quality conformance.- Quality conformance inspection upon completed clocks shall be conducted. A sample number of clocks selected in accordance with 4.4.1 shall be subjected to the examinations and tests specified in 4.5 through 4.6.4.

4.4.1 Sampling for quality conformance inspection.-

4.4.1.1 Inspection lot.- All clocks offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.4.1.2 Sampling for visual and dimensional examination.- A random sample of clocks shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105. Acceptable Quality Level (AQL) shall be equal to 1.5 percent defective.

4.4.1.3 Sampling for tests.- A random sample of clocks shall be selected from each inspection lot in accordance with inspection level L6 of the table for small sample inspection of MIL-STD-105. All clocks in the sample shall be subjected to the examination and tests specified in 4.5 and 4.6. If any clock fails any test, such failure shall be cause for rejection of the lot.

4.5 Examination.-

4.5.1 Visual and dimensional examination.- Each clock selected in accordance with 4.4.1.2 shall be visually and dimensionally examined to verify compliance with this specification. Any clock in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective clocks in any sample exceeds the acceptance number for that sample, it shall be cause for rejection of the lot represented by the sample. Defects are classified as shown in table I.

4.6 Test procedures.-

4.6.1 Illumination test.- The illumination of the assembled clocks shall be determined photoelectrically. The photoelectric device shall have been calibrated, not more than 6 months prior to this test, by a brightness standard certified for brightness by a laboratory satisfactory to the Government. All clocks to be tested shall be energized from a 115-volt, 60-cycle, single phase A. C. power source. With the dimming rheostat control turned fully clockwise for maximum light intensity, measure and record at least three readings of the brightness level over the entire surface of each lamp (dial and hands) using a photoelectric device. Repeat this procedure at reduced voltages, that is two-thirds and one-third of rated lamp voltage. Record the voltage and brightness levels at each position. Compare the highest

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:27Z
Check the source to verify that this is the current version before use.

brightness value to the lowest brightness value for each observation. If any ratio of brightness levels so obtained from each clock exceeds the requirements of 3.4.6.1, that clock shall be declared defective.

Table I - Classification of defects in accordance with MIL-STD-105

Categories	Defects
Critical:	
1	High voltage circuit not properly insulated to prevent shock hazard.
Major:	
101	Clock will not operate.
102	Label or proof of conformance to Underwriters' Laboratories requirements not furnished.
103	Time indication other than specified.
104	Dial and hands not properly illuminated.
105	Illumination control not provided; knob is loose or missing; not located properly.
106	Case material other than that specified; not proper color; method of mounting not as specified.
107	Manual reset control not located as specified; cable not located as specified.
108	Case back not as specified; mounting brackets not located as specified; warning label missing or not located as specified.
109	Dial cover broken; poor quality; not clear and free of defects.
110	Method of illumination not as specified; lamps not properly insulated; not designed for proper operating voltage; slip rings do not provide positive and smooth contact.
111	Dial markings not as specified; missing; incorrect size; not legible; dial scratched or damaged; not flat.
112	Hands bent; not of proper size as specified; rub against dial; hands will not move or are obstructed.
113	Transformer not as specified; secondary voltage not within specified limits; power input connection to transformer not as specified.
114	Motor not as specified; power requirement not within specified limits.
115	Cable not as specified; plug not as specified.
116	Instruction sheet missing; information not complete.
Minor	
201	Length of cable not as specified; not properly supported; abraded.
202	Current limiting device not provided.

4.6.2 Vibration test. - Each assembled clock shall be vibrated successively over the range of applied frequencies specified in table II in three mutually perpendicular directions; two directions of vibration shall lie in the plane of the clock. The frequency of applied vibration shall be varied uniformly from 2 to 25 cycles per second (c.p.s.) at a rate of approximately 1 c.p.s. per minute. The clock shall be clamped, in a normal vertical mounting position, on a vibrating table that can be controlled within 10 percent of the specified frequencies and amplitudes.

4.6.2.1 The clock shall be vibrated for at least 30 minutes in each of three directions specified in 4.6.2. Shock mounts shall not be blocked or removed. The clock shall show no evidence of damage affecting serviceability as a result of this test.

4.6.3 Temperature test. - The assembled clock shall be operated, without mechanical or electrical damage, under the temperature cycle specified in table III.

Source: https://assist.dla.mil -- Downloaded: 202610-19T16:27Z
Check the source to verify that this is the currentversion before use.

Table II - Amplitudes of vibration.

Frequency of Vibration	Amplitude (one-half total extension)	
	Horizontal	Vertical
c.p.s.	Inch	Inch
2 to 10	0.060	0.060
10 to 33	0.030	0.030

TABLE III - Temperature cycle.

Hours	Temperature	Humidity
1	75°F (24°C.)	95 percent
2	40°F (5°C.)	95 percent
5	149°F (65°C.)	95 percent

Any clock failing to pass this test shall be declared defective.

4.6.4 Dielectric strength test.- The dielectric strength between the electrical cirucit and ground of each assembled clock shall be tested with a closely sinusoidal source of 60 cycles A. C. having a capacity of at least 1 kilowatt. Root mean square (r.m.s.) values of test voltages shall be as specified in table IV.

Table IV - Dielectric test voltages.

Circuit voltage of clock to be tested	R.m.s. value of dielectric test voltage
120 and less than 240	1200
240 to 480	1500
above 480	Twice rated voltage plus 1000

The test voltage shall be raised gradually to the specified value and shall be held at the specified value for 5 seconds plus or minus 1 second. Any clock failing to pass this test shall be declared defective.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.-

5.1.1 Electric internally illuminated clocks.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early installation, and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier, at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number, contract or order number, contractor's name, and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, and packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.1).

5.2.1 The following provides the various levels for for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.1).

5.2.1.1 Preservation and packaging, packing, and marking.- Clocks shall be preserved and packaged level A or C, packed level A or B, and marked in accordance with PPP-T-360. For level A preservation and packaging, processing shall be in accordance with group 4 of PPP-T-360.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity required.
(c) Quantity of preproduction models required (see 3.1).
(d) Requirements and levels of preservation, packaging, packing and marking for domestic shipment and storage or overseas shipment if other than specified in 5.1 (see 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:27Z
Check the source to verify that this is the current version before use.

Notice.- When Government drawings, specifications, or other data are used for any purpose than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 6645-N011Sh)

Source: https://assist.dla.mil -- Downloaded: 202610-19T16:27Z
Check the source to verify that this is the current version before use.